



Deutsche Bank
Research

Q3 2026 Corporate Bank Quarterly

A guide to Macro, Markets, Geopolitics & Tech for CFOs and Corporates

July 13, 2026

**Marion Laboure | Camilla Siazon | Luke Templeman |
Adrian Cox | Helen Belopolsky | Miha Hribernik | Jim Reid**

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Introduction: Macro, Markets, Geopolitics & Tech

A guide for CFOs & Corporates



We're pleased to share our July 2026 Guide for CFOs and Corporates, providing essential insights into macroeconomics, markets, geopolitics and technology. This report offers critical analysis and perspectives to help corporate clients navigate today's complex economic environment.

Key highlights:

- **Macroeconomic outlook:** Resumption of US-Iran attacks in the Strait has reignited fears of an oil-driven energy shock. So far, the global economy has proved to be resilient.
- **Market analysis:** Corporate earnings growth is expected to be strong and credit spreads have proved resilient to several shocks. Brent crude averages \$80/bbl in 2026 (\$75 in Q3).
- **Geopolitics:** Corporates continue to cite geopolitics as a key risk, although other 'normal' risks are high priorities. Q3/Q4 events to watch: expiry of US global 10% Section 122 tariffs in Jul, expiration of original US-Iran ceasefire in Aug, US midterms in Nov.
- **Artificial Intelligence:** AI adoption is increasing, but corporates cite challenges in scaling across business functions. Disruption worries are relatively latent. The costs around AI capex demand are shifting into the spotlight.
- **Monetary policy:** The energy shock is influencing central bank trajectories, with both the Fed and ECB expected to hike rates. Momentum around policymakers considering tokenization of assets is growing.

Upcoming key macro events for rest of 2026

07 \ July			08 \ August			09 \ September		
01	IE	Ireland holds EU Council presidency	Early	CH	China Politburo meeting to set H2 economic policy direction	06-20	DE	German regional elections
07-08		NATO summit in Ankara				08	US	UNGA convenes in New York, starts next UN Secretary-General selection process
17	UK	Andy Burnham expected to become PM	Mid	US	60-day US-Iran ceasefire extension expires	09-10	EZ	ECB decision
22-23	EZ	ECB decision				12-13	IN	India hosts BRICS Summit
24	US	US global 10% Section 122 tariffs expire	29	IS	Iceland referendum on EU accession talks	13	SE	Swedish general election
28-29	US	Federal Reserve meeting				15-16*	US	Federal Reserve meeting
30	UK	Bank of England decision				17	UK	Bank of England decision
30-31	JN	Bank of Japan decision				17-18	JN	Bank of Japan decision
Late	CH	China Politburo meeting to set H2 economic policy direction				18-20	RU	Russian legislative election
						24	CH	President Xi expected to visit US after May China-US Summit
						27	FR	France Senate election
						30	US	Fiscal year ends- multiple agencies' funding expires

10 \ October			11 \ November			12 \ December		
04 & 25	BR	Brazilian general election	03	US	US midterm elections (House & Senate)	-	CH	China Central Economic Work Conference
27-28	US	Federal Reserve meeting	05	UK	Bank of England decision	08-09*	US	Federal Reserve meeting
28-29	EZ	ECB decision	09-20	COP31	COP31 in Türkiye	14-15	G20	G20 summit in US
29-30	JN	Bank of Japan decision	18-19	CH	APEC Economic Leaders' meeting in China	16-17	EZ	ECB decision
						17	UK	Bank of England decision
						17-18	JN	Bank of Japan decision

DB forecasts



GDP Growth (%)				
	2024	2025	2026F	2027F
Global	3.5	3.5	3.1	3.3
US	2.8	2.1	2.3	2.2
Eurozone	0.9	1.5	0.5	1.1
Germany	-0.5	0.2	0.5	1.3
France	1.4	0.9	0.5	0.9
Italy	0.6	0.7	0.4	0.7
Spain	3.5	2.8	1.8	1.8
Japan	-0.2	1.1	0.7	0.7
UK	1.0	1.3	1.0	1.2
China	5.0	5.0	4.7	4.5
India	7.2	7.5	6.7	7.2
Russia	4.9	1.0	0.6	1.3
Asia (ex Japan)	5.4	5.5	5.2	5.1
EEMEA	3.5	3.1	2.4	3.4
LatAm	2.6	1.8	1.9	1.9
EM	4.6	4.5	4.2	4.4

- *CPI (%) forecasts are period averages. **Germany CPI local definition taken.
EEMEA: Czech Rep., Israel, Egypt, Hungary, Poland, Russia, Saudi Arabia, South Africa, Turkey, UAE
- LATAM (ex Venezuela): Argentina, Brazil, Chile, Colombia, Mexico, Peru
- ASIA (ex Japan): China, HK, India, Indonesia, Korea, Malaysia, Philippines, Singapore, Sri Lanka, Taiwan, Thailand, Vietnam
- Current prices as of July 10, 2026.
- *China rates are 7d OMO rates.
- **Gold, Brent and WTI forecasts are for quarterly periods.

CPI Inflation, YoY* (%)				
	2025	2026F	2027F	2028F
US	2.7	3.2	2.2	2.6
Eurozone	2.1	2.8	2.3	2.2
Germany**	2.3	2.9	2.7	2.4
Japan	3.2	2.1	2.6	2.1
UK	3.4	3.2	2.7	2.1
China	0.1	1.3	1.6	-
India	2.2	4.6	4.2	
Russia	8.7	5.5	5.0	

Central Bank policy rate (%)				
	Current	Dec 26F	Jun 27F	Dec 27F
US	3.625	4.125	4.125	4.125
Eurozone	2.25	2.50	2.50	2.50
UK	3.75	3.75	3.50	3.25
Japan	1.00	1.25	1.75	1.75
China*	1.40	1.40	1.40	1.40
India	5.25	5.75	6.25	6.25

Key market metrics				
	Current	Dec 26F	Jun 27F	Dec 27F
US 10Y Yield (%)	4.56	4.80	4.75	-
EUR 10Y Yield (%)	3.06	3.20	3.20	3.20
EUR/USD	1.14	1.20	-	1.25
USD/JPY	162	150	-	140
USD/CNY	6.78	6.55	6.50	6.30
S&P 500	7,575	8,000	-	-
**Oil WTI (USD/bbl)	71	67	67	67
**Oil Brent (USD/bbl)	76	70	70	70
**Gold	4,119	4,800	5,200	5,400

Source: Deutsche Bank Research, Bloomberg Finance LP.



01. Macroeconomics



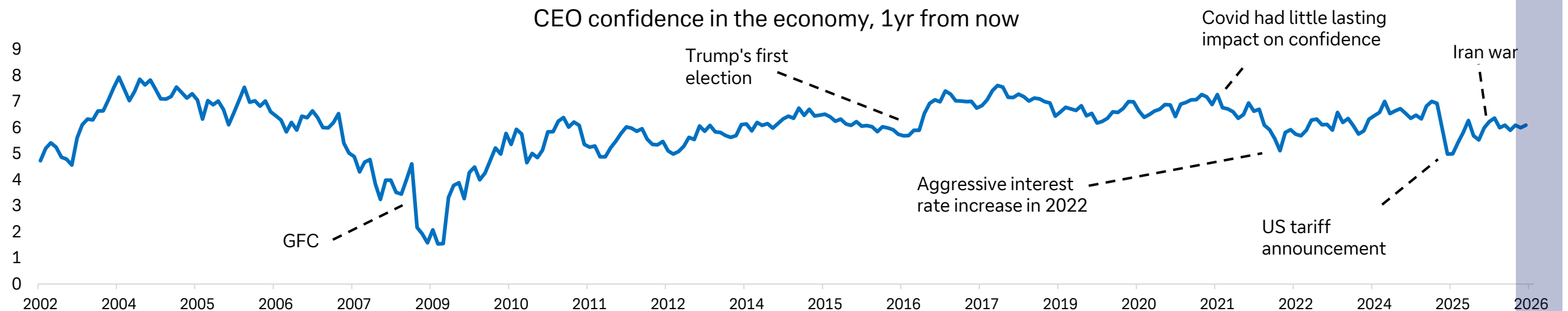
2026 Global growth

The situation in the Middle East conflict adds uncertainty, but growth remains far more resilient than pre-war forecasts assumed for a Strait of Hormuz closure



View	Rationale	View	Rationale
 Growth at 3.1% in 2026 and 3.3% in 2027.	— Main risk is if the war continues/worsens, oil market impact from prolonged SoH closure.	 Growth at 4.7% in 2026, 4.5% in 2027.	— Softer domestic demand and policy adjustments offset by strong trade and anticipated reflation.
 2.3% GDP in 2026 and 2.2% in 2027.	— Growth to stay resilient in 2026, supported by easy credit conditions, fiscal policy, and AI-driven capex. A more substantial oil shock can weaken real disposable income and tighten financial conditions.	EM 4.2% growth for the EM region in 2026, 4.4% in 2027.	— In Asia, growth momentum is resilient, with fiscal balance sheets actively working to dampen the transmission of the energy-price shock.
 Euro area growth 0.5% in 2026 and 1.1% in 2027. German GDP at 0.5% in 2026 and 1.3% in 2027.	— Europe remains materially impacted by the Middle East crisis and energy shock.		— CEEMEA's economic growth proved relatively durable through the conflict, though 2026 growth was cut to 2.4%. Sharp reversal in oil prices during MOU has lifted inflation and allowed financial conditions to normalize.

CEO confidence edged back up in Q2, as a ceasefire was announced – Q3 might be a different story



Source: Deutsche Bank Research, Bloomberg Finance LP.

Three key risks for the global economy

Collapse of the Iran ceasefire is renewing stagflation concerns, while cybersecurity risks have become a key concern for corporates

1. Collapse of Iran ceasefire

- Renewed flare-ups in the Middle East are driving oil prices higher again, compounded by energy infrastructure attacks between Russia and Ukraine. Corporates remain concerned with economic conditions [vs six months ago](#).
- Long term, however, the global economy seems more resilient to geopolitical shocks.

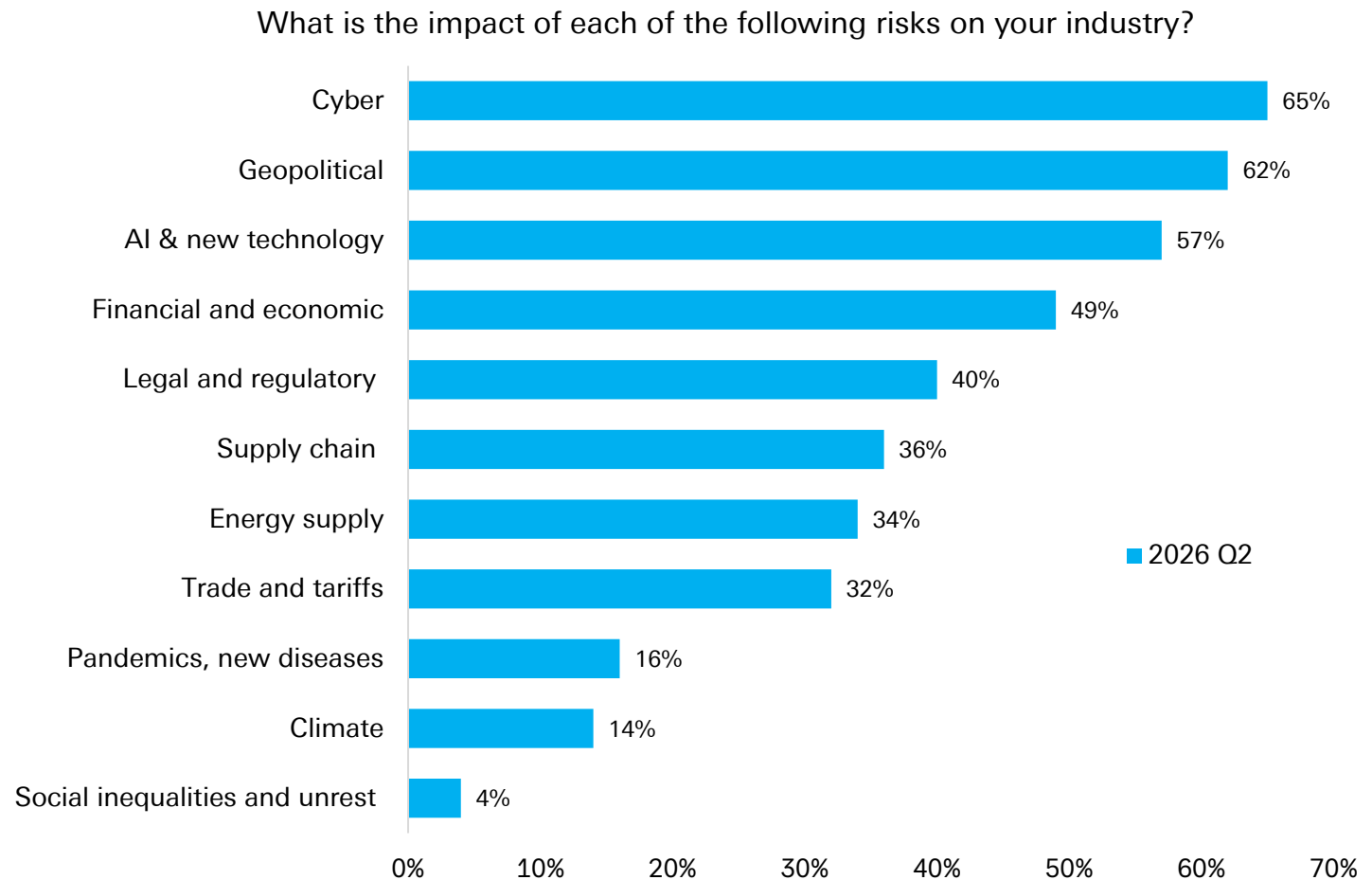
2. AI disruption

- Concerns that AI will disrupt the labor force have yet to be borne out by the data. CEOs see little impact on jobs in the near term, while companies struggle to diffuse adoption.
- Central banks are becoming more attuned to AI as a source of upwards price pressure.

3. Hawkish central bank trajectories

- Expectations for higher US rates will grow as markets absorb the apparent demise of the ceasefire. Rates expectations had already decoupled from oil prices even during the ceasefire.
- Expectations for ECB rate hikes have also shifted due to the energy shock.

Cyber risk has emerged as the top concern among CEOs



Source: Deutsche Bank Research, The Conference Board Measure of CEO Confidence in collaboration with The Business Council

Central banks

The inflation shock has altered central bank trajectories



Fed

- +25bps hike in Sep and Dec, bringing Fed funds rate to 4.1% YE, on hold in 2027. 50bps of rate cuts pencilled in for H1 2028.
- Core PCE sticky at 3.2% in 2026 and 2.5% in 2027.



BoE

- On hold at 3.75% through 2026, with rate cuts to resume from spring next year.



BoJ

- Quarterly +25bp rate hikes starting in Oct, 1.75% policy rate by Apr 2027.



PBoC

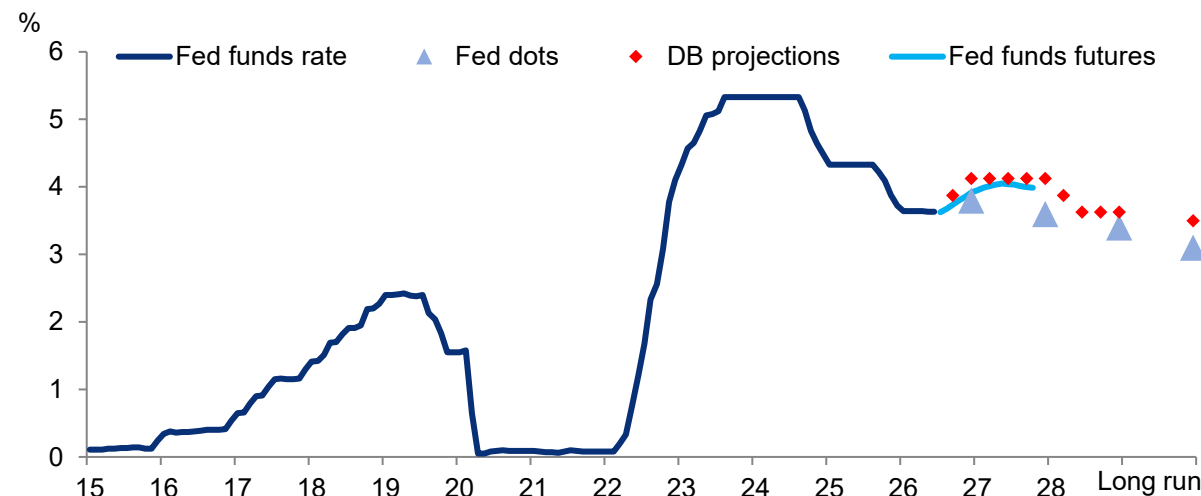
- Hold the 7-day reverse repo at 1.40% through 2026.



ECB

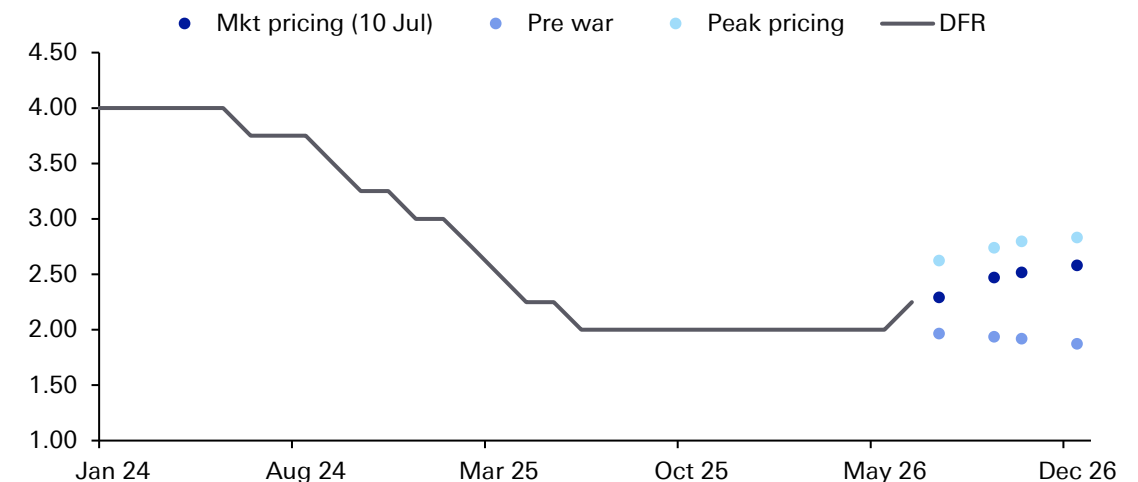
- A further +25bp hike in Sep to 2.50% year-end.
- ECB staff “baseline scenario” consistent with 2.75% rates, ‘milder’ scenario consistent with 2.50% rates.

Fed: a December hike if fully priced



Source: FRB, Bloomberg Finance LP, Haver Analytics, Deutsche Bank Research. Updated July 2 2026.

ECB: 2.50% YE



Source: Deutsche Bank Research, ECB. Updated July 10 2026.



02. Geopolitics



Midterm Moderation: Will Trump Turn Dove?



Trump has a strong incentive not to shock markets or stoke inflation ahead of the vote in November. But markets should expect more policy volatility, not less, as we move into the second half of Trump's term. An administration looking to demonstrate strength through quick wins may pursue other issues more aggressively. Investors should watch rising US pressure on Cuba and the political relationship with Europe as possible sources of nearer-term movement.

The **July expiry of Section 122 tariffs** and their likely replacement by Section 301 levies on dozens of trading partners should keep pressure on countries to strike bilateral deals favourable to the US, spanning trade, investment, industrial and technology concessions. In addition, Section 232 tariffs could expand into new categories.

While **EU-US trade relations are more stable**, there remains a tail risk that a transatlantic security or sovereignty dispute (e.g. renewed American claims on Greenland) spills over into the economic relationship. At the 7-8 July NATO summit, Trump reinforced his expectations that European allies carry more of the security burden and his disappointment in their lack of support for US efforts in Iran. Trump also reiterated his continued interest in acquiring Greenland.

Trump will likely renew his push for a ceasefire in the **Russia-Ukraine war**. In early July, Trump suggested “we’re getting much closer than people realize.” While a ceasefire this year is far from guaranteed, any deal brokered by the Trump administration could offer Russia sanctions relief, access to frozen assets and an expectation of an accelerated Ukrainian accession to the EU.

What happens after the midterms actually matters more. **Trump may become a “foreign policy president”**, focused on cementing his legacy. This will include: embedding existing tariff and investment deals with key US partners; deepening US leadership in the Americas; creating the conditions for a pull-back of American resources from Europe; and broadening US efforts to reduce rare-earths and critical-minerals dependence on China.



Source: [Library of Congress](#) on [Unsplash](#)

Other geopolitical flashpoints



Key topics to monitor during 2026-Q3

- **Iran:** The 60-day limit for US-Iranian MOU (focusing on Iran's nuclear programme and deal implementation) will expire on 20 AUG but negotiations will almost certainly run past the initial 60 days. A significant escalation on over the past week saw the US launch strikes on hundreds of targets across Iran and reimpose sanctions on Iranian oil sales in retaliation for Iran striking ships in the Strait of Hormuz. Iran has fired a number of waves of strikes against Gulf nations. While Trump has suggested that the ceasefire is over, we retain our assessment that both parties are incentivized to find a near-term resolution as kinetic attacks will not drive a resolution. That said, we should prepare for more volatility in the coming months and a persistent threat to the Strait of Hormuz. The Middle East will likely remain a focus for the rest of Trump's term, even if markets move on.
- **China-US Relations:** Post-summit strategic stability could deliver tactical trade and investment gains and slow the pace of escalation into late 2026. But corporates and investors should still prepare for a start-stop cycle of engagement and restriction as technology, tariffs and economic security remain central to national security agendas in Beijing and Washington. A key watchpoint will be Xi's expected visit to the US on 24 September, just over a month before the US midterm election set to be dominated by voter cost-of-living concerns. This could incentivize the Trump administration to seek an economic agreement at the summit. An extension of last **October's Busan agreement** and an operationalization of the potentially commerce-boosting **US-China Board of Trade** – which China hopes could include a ten-fold increase in tariff-free trade – are possible deliverables.
- **What next for Europe?** An unresolved war in Ukraine, greater geopolitical fragmentation, and a less predictable transatlantic relationship. At the same time, the geopolitical landscape is presenting myriad opportunities – increased defence investment and a chance to build more strategic autonomy. There is a growing recognition that the EU needs to overcome its often slow, consensus-driven decision-making processes to respond more effectively to evolving geopolitical dynamics. The key question will be whether Europe can convert these pressures into coordinated action, or whether internal divisions stymie real progress.

For more on geopolitical flashpoints to watch out for, read our recent publications here: [Geopolitical Outlook 2026: At the Midpoint](#), [This Month in Geopolitics: July 2026](#), [Geopolitics beyond Iran](#), [The Dragon and the Eagle: Decoding China-US Ties After Beijing](#)

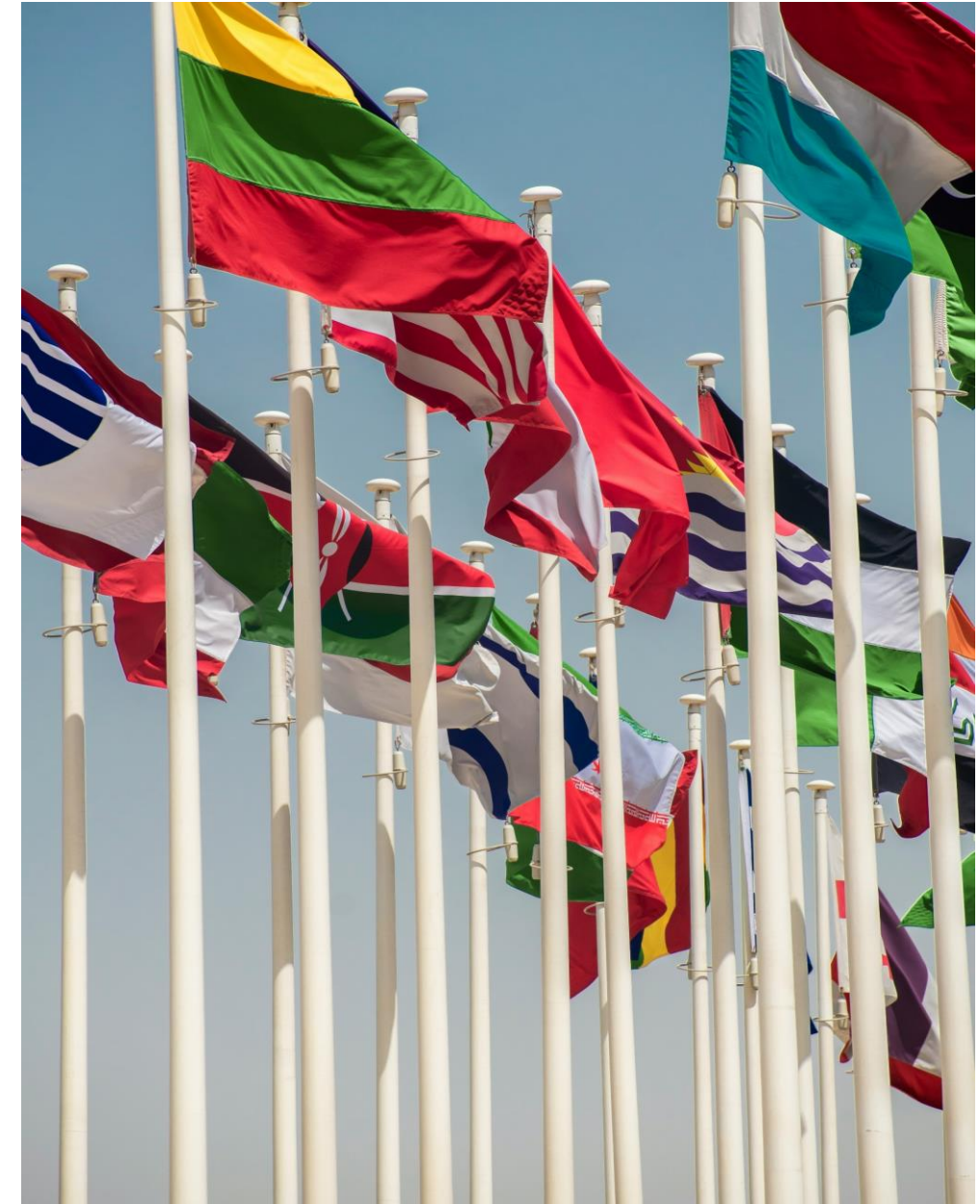


Source: IStock ugurhan

Selected geopolitical events in July - September



- **Mid-July:** **UK:** Andy Burnham expected to formally become Prime Minister.
- **24 July:** The **US** global 10% Section 122 tariffs are set to expire.
- **Late July/early Aug:** The **Chinese** Politburo meeting will likely set the economic policy direction for the rest of the year.
- **Mid-August:** **US/Iran:** The 60-day ceasefire extension between the US and Iran is set to expire.
- **29 Aug:** **Iceland** referendum on EU accession talks.
- **8 Sep:** The UN General Assembly in **New York** will convene and initiate the process to choose the next UN Secretary-General.
- **12-13 Sep:** **India** will host the BRICS Summit in New Delhi.
- **13 Sep:** **German** state and local elections.
- **13 Sep:** **Swedish** general election.
- **18-20 Sep:** **Russian** legislative election.
- **24 Sep:** **China's** President Xi is expected to visit the United States following the China-US Summit in Beijing in May.
- **30 Sep:** Key **US** government funding deadline. Failure to pass appropriations or a stopgap measure could trigger a government shutdown.



Source: [aboodi vesakaran](#) on [Unsplash](#)



03. Markets



Commodity markets

Oil was trading around \$73/bbl, but prices are now rising back towards \$80/bbl. Several transport- and materials-exposed companies have pointed to 2027 for normalization (see [here](#)).



Oil: Under the ceasefire, we had placed Brent at \$75/bbl (Q3'26) and \$70/bbl (Q4'26).

- Oil prices fell faster than expected during the MOU-interim, although outlook is uncertain for now with the reignition of Strait conflict.
- Crisis has revealed need for broader reserves beyond crude oil and gas, where disruption can rapidly cascade through supply chains and consumer prices (i.e. refined fuels, petrochemicals, fertilizers, food, metals).
- Companies should now be expected to maintain larger inventories, build in more redundancy, spend more on insurance, hedging, and transport, and accept duplicated routes and suppliers.

Gold: revised base case to reach \$4,800/oz in Q4

- Tighter Fed monetary policy and resilient US macro data has pushed gold prices lower.

Latest oil price forecasts

USD	2025	Q3 2026	Q4 2026	2026	2027	2028
WTI (bbl)	64.9	72.0	67.0	76.1	67.0	73.3
% change		-23.4%	-17.3%	-10.5%	-11.8%	-2.8%
Brent (bbl)	68.20	75.0	70.0	80.0	70.0	76.4
% change		-24.2%	-18.6%	-11.1%	-12.5%	-2.7%

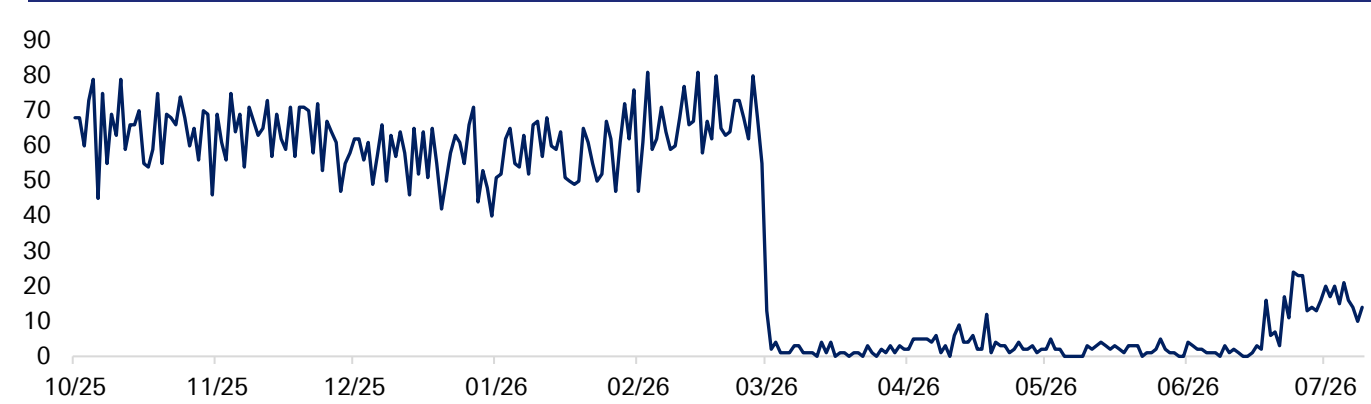
Reports: [Hsueh On Oil: Swinging to surplus - Deutsche Bank Research](#) and [Precious Special Report: Hawks drive out bulls - Deutsche Bank Research](#)

Brent crude oil prices (\$/bbl) have gone from around \$100/bbl to just \$73/bbl during MOU



Source: Deutsche Bank Research, Bloomberg Finance LP

The number of vessels going through the Strait of Hormuz is still at a fraction of its pre-conflict level



Source: Deutsche Bank Research, Bloomberg Finance LP.

Rates & Credit markets

Despite higher rates, financial conditions are relatively accommodative, with credit spreads remaining tighter than expected. Potential area for concern lies in private credit



Private credit is pricing in more stress now that full Q1 data is in. There is potential for greater AI-disruption stress on software and other adjacent subsectors.

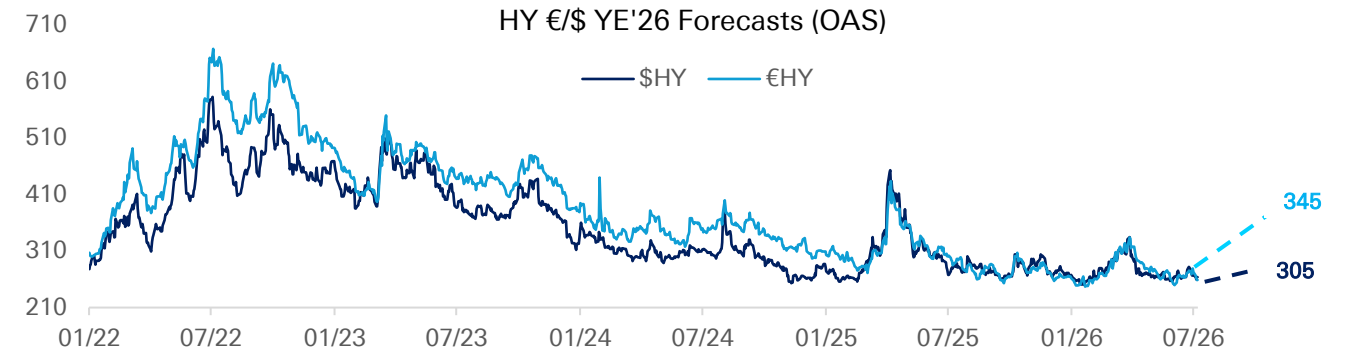
- Software spreads are near Covid (March 2020) levels, but ex-software spreads are trading much better. Market is pricing software as an isolated issue, although there is a risk the stress will spread more broadly over the next 6-12 months.
- Credit-spread forecasts slightly lower throughout 2026, with \$IG 82bps; \$HY 305bps. €IG 95bps; €HY 345bps.

Raised our YE-2026 targets for the 10yr US Treasury (4.80%) and 10yr Bund (3.20%)

- Rationale: Expectations of more Fed hikes are now baked into our forecast, while front loading of ECB hikes in response to the energy shock has pushed our yield forecasts higher across the curve.
- Downside risks: AI tails from either a bubble or significant impact on the market, or a concrete end to the Iran War leading to materially lower oil prices.
- Upside risks: Demand for additional fiscal and capex globally or a waning in foreign demand for US Treasuries that further lifts term premia.

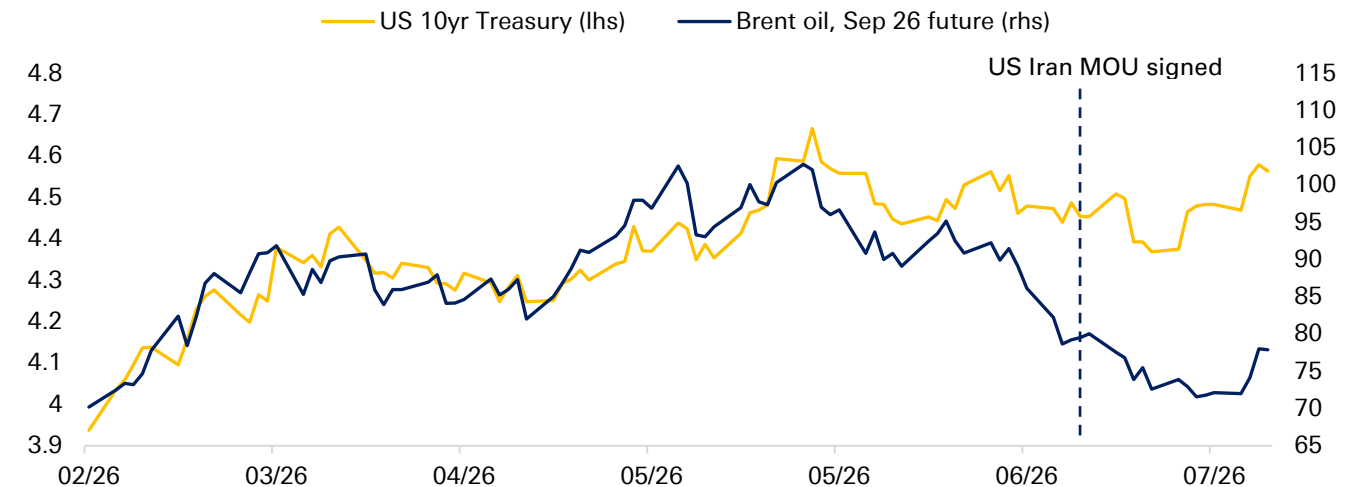
Reports: [Global Fixed Income Weekly: Strategy Update - Deutsche Bank Research](#) and [Fixed Income Blog: EUR Rates Forecast revision - Bund to 3.2% - Deutsche Bank Research](#) and [IG & HY Strategy: Global Default Monitor: Software woes, BDC selling putting pressure on Loans - Deutsche Bank Research](#)

Outlook until year-end calls for more dispersion and some spread widening



Source: Deutsche Bank Research, Bloomberg Finance LP.

The ten year treasury yield initially traded in a tight correlation with oil prices, but that's completely opened up in the last month. Yields have stayed high even during the ceasefire



Source: Deutsche Bank Research, Bloomberg Finance LP.

Foreign exchange markets

Hawkish FOMC is supporting the dollar

Hard to argue for significant dollar weakness

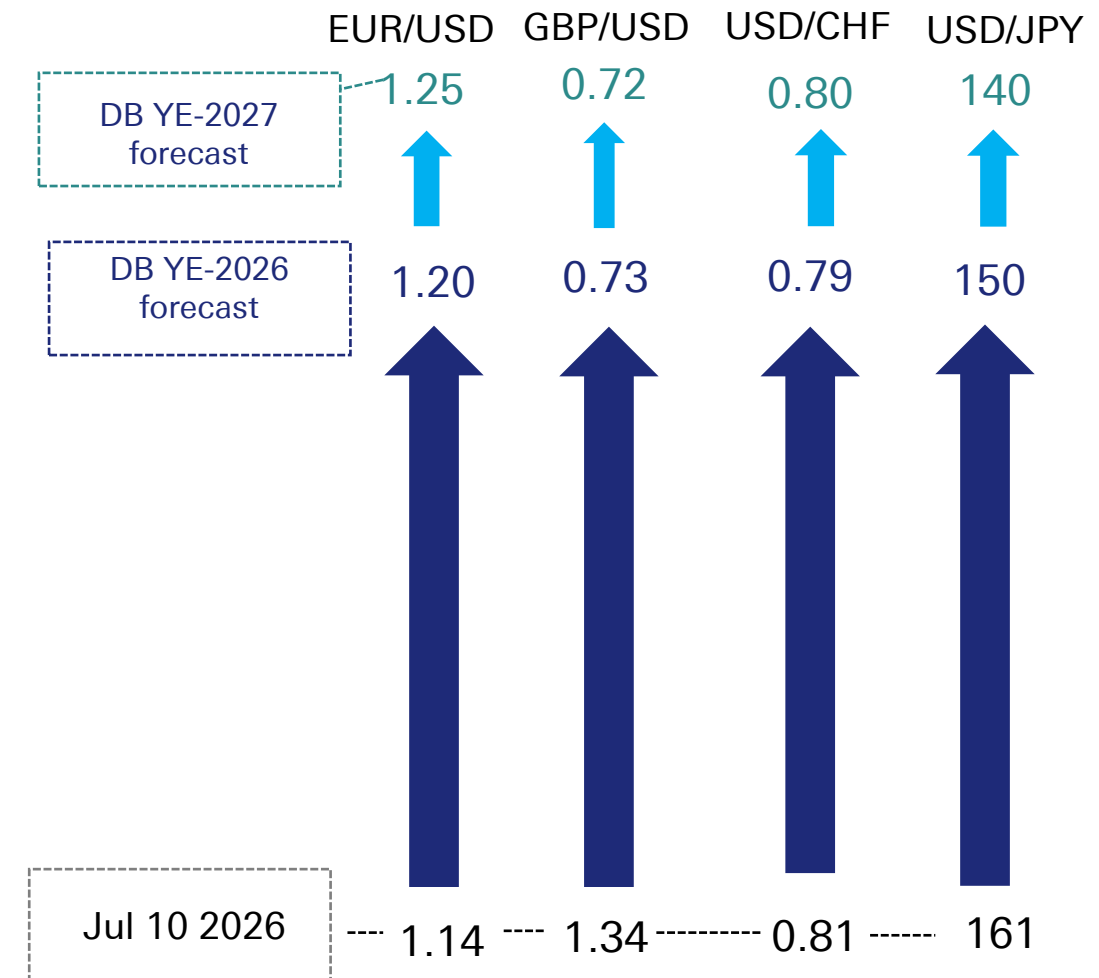
- The dollar has modestly benefited from rising energy prices, while equity-led inflows into the US have also started to recover.
- Near-term, the second-order effects of the Iran War is dollar supportive.
- However, the level of front-end US Treasury yields, the interest-rate differential with the rest of the world and relative shape of the US yield curve are not sending dollar bullish signals.

Currency outlook

- **EUR/USD:** 1.20 by YE (prevs 1.25). Escalation in Iran and more hawkish repricing of the Fed could push EUR/USD below 1.15.
- **EUR/JPY:** Hawkish signals from BoJ officials, firm recent data, and potential for repatriation sees the yen rising against the Euro. Despite yen relatively flat against the Euro since our [May blueprint](#), we still expect the yen to strengthen in H2, with two potential avenues: either the BoJ hikes rates to 2% more quickly than expected, or the government undertakes a more concerted effort to shift domestic investment back onshore.
- **USD/CNH:** Renminbi's appreciation will continue. Domestic reflation, strong real-GDP growth, surpluses, and a greater use of RMB in cross-border transactions/internationalisation should all act as significant tailwinds.

Read: [FX Blog: one thing - Deutsche Bank Research](#)

Currency movements and DB forecasts for YE-2026 and 2027



Source: Deutsche Bank Research, Bloomberg Finance LP.



04. Technology



Artificial intelligence

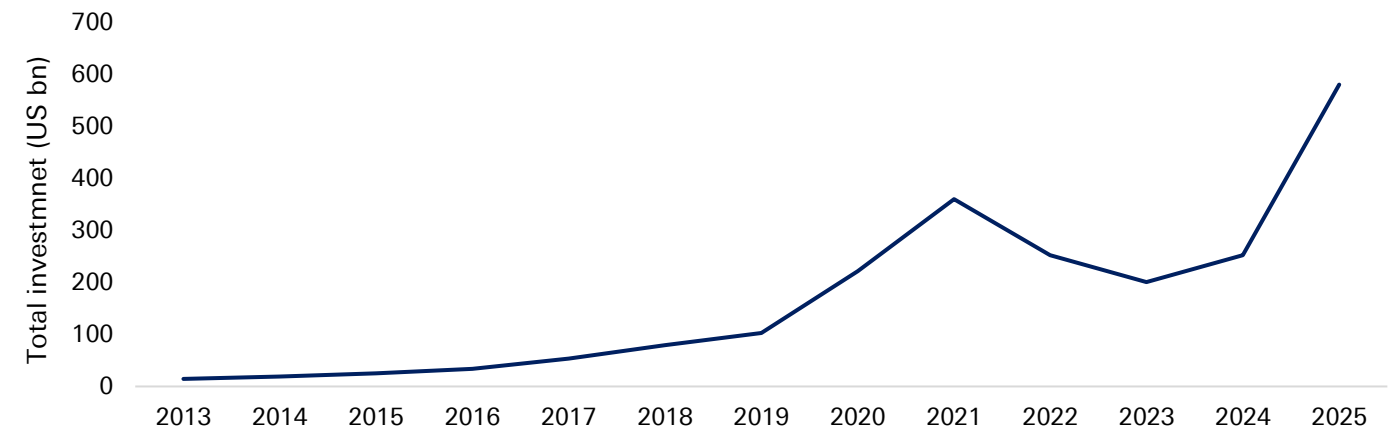
Companies are committing growing sums of capital to strengthen their AI capabilities



2025 was a breakout year for corporate AI investment

- Global AI corporate investment hit \$582bn in 2025, +130% from 2024, mostly driven by Gen AI. (Source: [McKinsey/Stanford](#)).
- Private financing (VC/PE) was the largest share of activity.
- The US continues to lead private AI investment; China's true total AI spending is understated, as government guidance funds deploy a significant amount.

Global corporate investment in AI

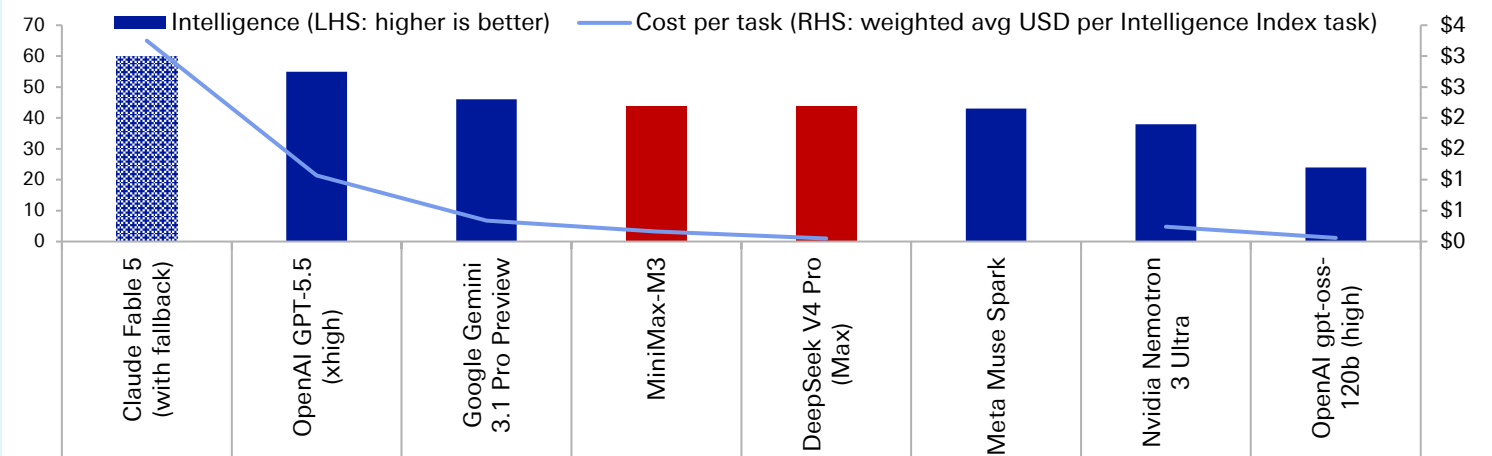


Source: Deutsche Bank Research, McKinsey/Stanford Artificial Intelligence Index.

The cost of tokens for frontier models is exponentially higher than still very intelligent mid-tier models. This is one of the biggest challenges for widespread company adoption.

The costs for AI are growing

- AI company revenues are rising at record pace – but so are compute and infrastructure costs.
- Frontier-model token costs specifically have become exponentially higher than still very intelligent mid-tier models. This is one of the biggest challenges for wide enterprise adoption.
- There is also evidence the AI-capex demand shock is feeding into inflation, most visibly in computer and semiconductor prices, specifically memory.



Source: Deutsche Bank Research, Artificial Analysis Intelligence Index. "Fallback" mechanism routes safety-flagged messages to Opus 4.8 model. Meta Muse Spark price not available. US models in blue; Chinese models in red

Artificial intelligence

Job displacement from AI does not seem to be a near-term (~2 years) concern



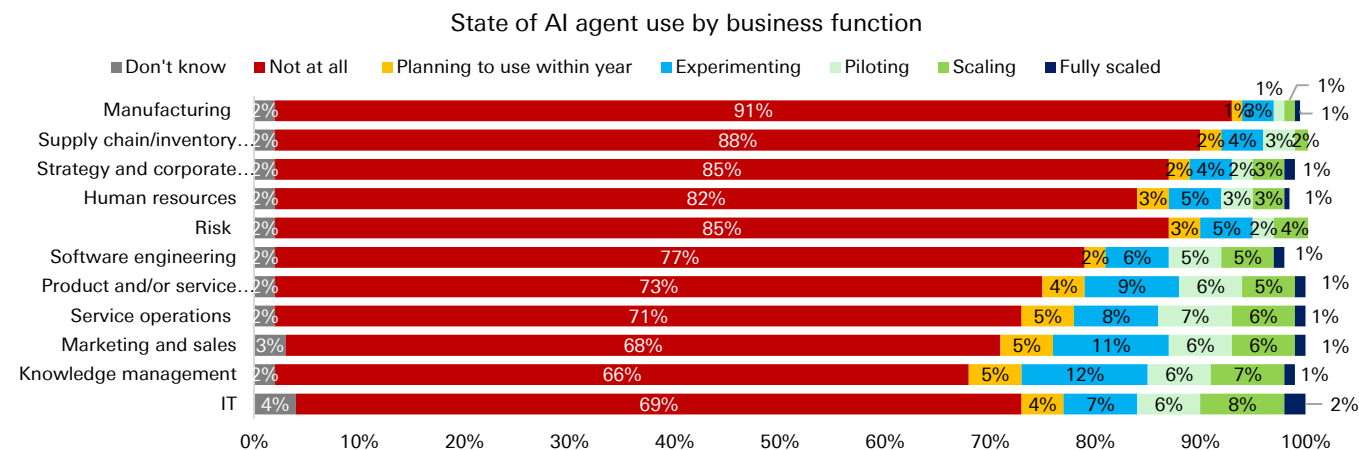
Scaling AI adoption remains a challenge

- 88% of firms now use AI in at least one business function (78% in 2024). Source: [McKinsey/Stanford AI index](#)
- Biggest y/y gains: Europe (+11pp to 91%) and China (+13pp to 88%) (McKinsey/Stanford).
- Usage is highest in knowledge management for business, legal, and professional services (58%), software engineering (58%), and IT (56%).
- But scaling adoption lags: <10% of firms are scaling AI agents across their firm.

AI job displacement not a near-term concern

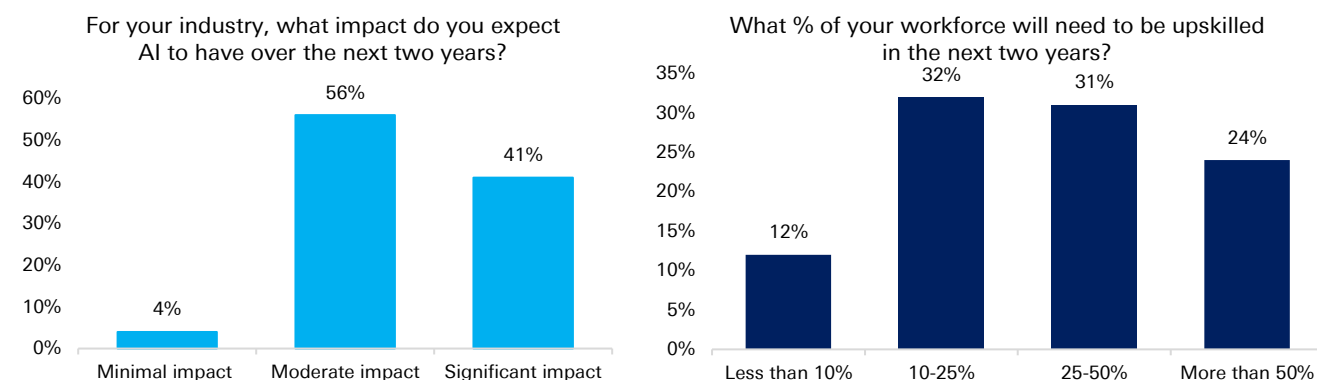
- 56% of CEOs expect AI to have only a moderate impact on their industries (Business Council and Conference Board).
- Evidence so far shows little relationship between US labour-market churn and AI adoption rate (see [here](#)).
- Yet, a third of firms expect AI to cut headcount in the coming year – larger firms more so. The largest share (43%) still expect little or no change. McKinsey/Stanford

Organizations are beginning to experiment with agentic AI, but diffusion is not yet widespread



Source: Deutsche Bank Research, McKinsey/Stanford Artificial Intelligence Index. Number of respondents = 1,933

Majority of CEOs don't think AI will fundamentally transform their firms, but most also anticipate upskilling their workforce in the short term



Source: Deutsche Bank Research, The Conference Board Measure of CEO Confidence in collaboration with the Business Council, Q2 2026. Left hand chart: Minimal impact (very few business processes will change; the industry remains largely unchanged); moderate impact (many business processes will change, but the industry will not fundamentally transform); Significant impact (AI will affect nearly every business process the industry will be fundamentally transformed.)



Digital assets

The push toward 24/7, digital, and programmable financial markets has brought tokenization into focus for policymakers and market infrastructure providers

Asset Tokenization

- Transferable tokenized assets on-chain total \$33bn (rwa.xyz).
- [The SEC is reportedly](#) moving to permit trading of tokenized stocks; ongoing DTCC progress on tokenized collateral infrastructure.
- Several US banks have announced a tokenized-deposits clearing network operated by The Clearing House, live from 2027 onwards, with settlement in central bank reserves.

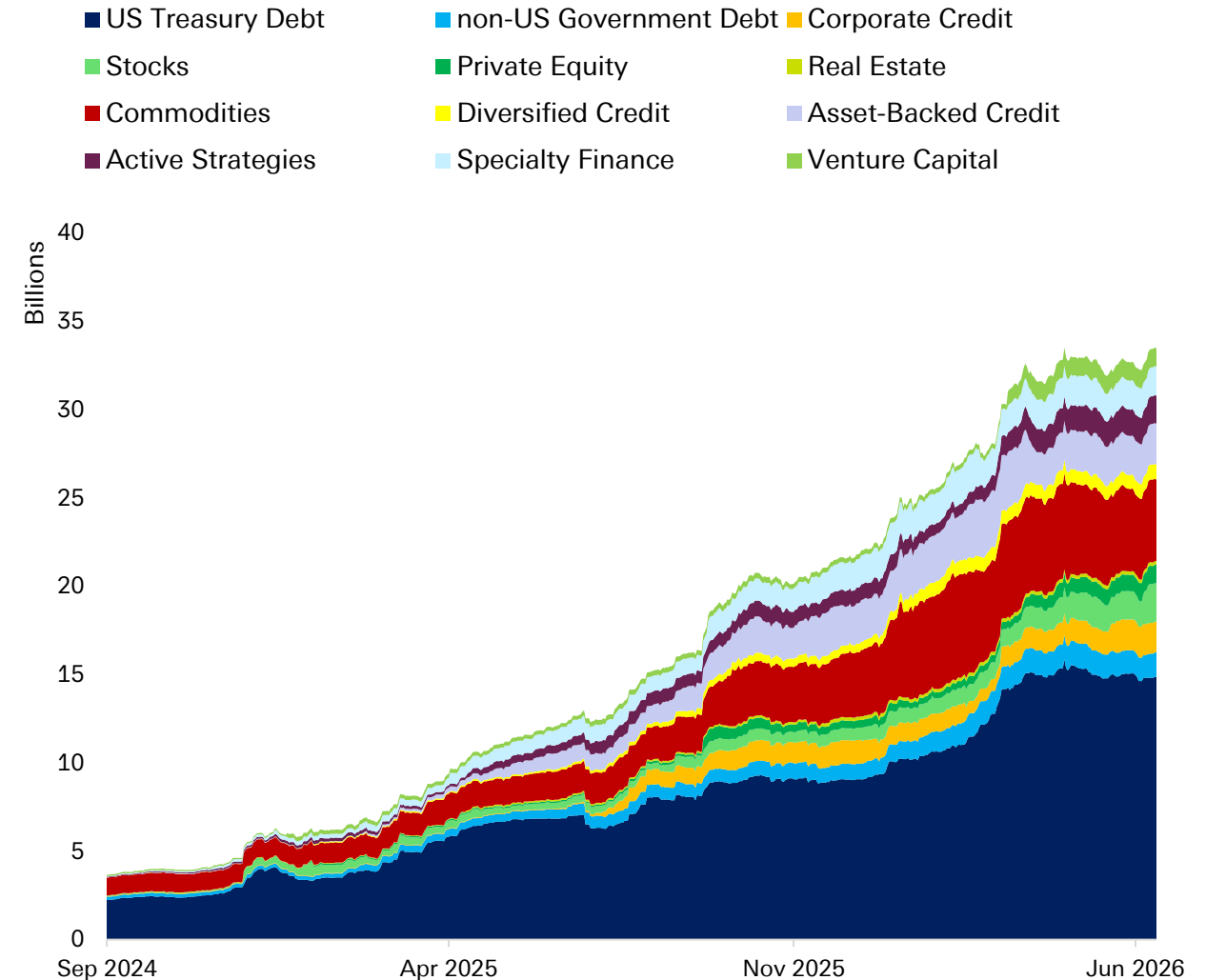
Stablecoins

- Adjusted stablecoin transaction volume is projected to reach [\\$719trn](#) by 2035 (Chainalysis).
- Stablecoin payment volumes could match Visa and Mastercard's off-chain transactions sometime between 2029-31.

CBDCs

- Wholesale: ECB's Project Pontes – a settlement solution for DLT-based transactions in central bank money – tracking for Sep 2026 launch.
- Retail: EP committee approved the digital euro framework June 23, final law targeted end-2026. Pilot mid-2027; first issuance possible 2029.

Tokenization is growing fast



Source: Deutsche Bank Research, RWA database. Data up to July 10.



Appendix 1

This material has been prepared by the Deutsche Bank Research Institute and is provided to you for general information purposes only. The Institute leverages the views, opinions, and research from Deutsche Bank Research, economists, strategists, and research analysts. Accordingly, you should assume that content in this document is based on or was previously published and provided to Deutsche Bank clients who may have already traded on the basis of it.

Any views or estimates expressed in this material reflect the current views of the author(s) and may differ from the views and estimates of Deutsche Bank AG, its affiliates, other Deutsche Bank personnel, and other materials published by Deutsche Bank. The content in this material is valid as of the date shown on the first page and may change without notice. Deutsche Bank has no obligation to provide any updates or changes to the information herein.

This material should not be used as a basis for trading securities or other financial products and should not be considered to be a recommendation or individual investment advice for any particular person. It does not constitute an offer, solicitation, or an invitation by or on behalf of Deutsche Bank to any person to buy or sell any security or financial instrument. Nothing in this material constitutes investment, legal, accounting or tax advice. Deutsche Bank engages in securities transactions, including on a proprietary basis, and may do so in a manner inconsistent with the views or information expressed in this material.

While information in this material has been obtained from sources believed to be reliable, neither Deutsche Bank AG nor any of its affiliates makes any representation or warranty, express or implied, as to the accuracy or completeness of the statements or any information contained in this material and therefore any liability is expressly disclaimed. This material is provided without any obligation, whether contractual or otherwise. Information regarding past transactions or performance is not indicative of future results.

In the U.S. this report is approved and/or distributed by Deutsche Bank Securities Inc., a member of FINRA. In Germany this information is approved and/or communicated by Deutsche Bank AG Frankfurt, licensed to carry on banking business and to provide financial services under the supervision of the European Central Bank (ECB) and the German Federal Financial Supervisory Authority (BaFin). In the United Kingdom this information is approved and/or communicated by Deutsche Bank AG, London Branch, a member of the London Stock Exchange, authorized by UK's Prudential Regulation Authority (PRA) and subject to limited regulation by the UK's Financial Conduct Authority (FCA) (under number 150018) and by the PRA. This information is distributed in Hong Kong by Deutsche Bank AG, Hong Kong Branch, in Korea by Deutsche Securities Korea Co. and in Singapore by Deutsche Bank AG, Singapore Branch. In Japan this information is approved and/or distributed by Deutsche Securities Limited, Tokyo Branch. In Australia, retail clients should obtain a copy of a Product Disclosure Statement (PDS) relating to any financial product referred to in this report and consider the PDS before making any decision about whether to acquire the product.

By accessing this material, you agree that its content may not be reproduced, distributed or published by any person for any purpose, in whole or part, without Deutsche Bank's prior written consent. You also agree that you shall not scrape, extract, download, upload, sell or offer for sale any of the content in this material, and you agree not to use, or cause to be used, any computerized or other manual or automated program or mechanism, tool, or process, including any scraper or spider robot, to access, extract, download, scrape, data mine, display, transmit, or publish, any of the content in this material.